Supplemental Material: Extensible Causality Does Not Guarantee Pure Quantum Dilations

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The main text distinguishes causal laboratory records from a pure quantum explanation of the process connecting the laboratories. This supplement gives the explicit local response protocols and the complete proof: a process witness excludes every finite pure dilation, while a common order weight and an arbitrary-dimensional positive operator identity establish causality for every finite experiment.

S1 Definitions and conventions

The original ports are qubits in the order A_I, A_O, B_I, B_O: Alice's input and output, then Bob's input and output. The subscripts I, O mean input and output. The Pauli matrices are $\sigma_x, \sigma_y, \sigma_z$ in the computational basis. A superscript on a Pauli matrix identifies its port; omitted ports carry the identity. Products on distinct ports denote tensor products in the stated order. Without port labels, the tensor factors are specified locally. I without a subscript denotes the identity on the specified factors, and I_T the identity on a space T. Input ancillas A' and B' can have arbitrary finite dimensions. Alice chooses a setting x and records an outcome a; Bob chooses y and records b. A process matrix W assigns probabilities

$$p(a,b|x,y) = \mathrm{Tr}\Big[(W \otimes \Omega^{A'B'})(M_{a|x}^{A_I A' A_O} \otimes N_{b|y}^{B_I B' B_O})\Big], \tag{S1}$$

where $p(a,b|x,y)$ is the joint conditional probability, $\Omega^{A'B'}$ the shared input density operator, and $M_{a|x}, N_{b|y}$ the Choi matrices of the local operations for the indicated settings and outcomes. Superscripts on these matrices list the spaces on which they act; factors are reordered as needed. Tr is the trace, $\otimes$ the tensor product, and $\succeq 0$ positive semidefiniteness; $U \preceq V$ means $V - U \succeq 0$. Thus $\Omega \succeq 0$ and $\mathrm{Tr}\,\Omega = 1$. Each event is completely positive (CP), and summing an instrument over outcomes gives a trace-preserving (TP) channel.

The Choi convention matters for complex operations. For a CP map $\mathcal{E}$, define its standard Choi matrix $J_{\mathcal{E}}$ and the matrix $M_{\mathcal{E}}$ used in Eq. (S1) by

$$J_{\mathcal{E}} = \sum_{i,j} |i\rangle\langle j| \otimes \mathcal{E}(|i\rangle\langle j|), \qquad M_{\mathcal{E}} = J_{\mathcal{E}}^T, \tag{S2}$$

where $|i\rangle, |j\rangle$ are input computational-basis vectors and the sum runs over that basis. The superscript T is the full transpose in the input–output product basis, $*$ denotes entrywise complex conjugation,

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and † below denotes the Hermitian adjoint. Equivalently, $M_{\mathcal{E}} = J_{\mathcal{E}}^{*}$, since $J_{\mathcal{E}}$ is Hermitian. Thus $M_{\mathcal{E}} \succeq 0$ and $\mathrm{Tr}_O\, M_{\mathcal{E}} = I_I$ for a channel, where Tr_O traces out the output and I_I is the input identity. Equation (S1) contains no additional transpose.

For an operator Q and a tensor factor T of dimension d_T, the replacement map ${}_TQ$ traces out T and reinserts the normalized identity I_T/d_T. Multiple labels indicate replacement of all listed factors. With trivial original global past and future, the bipartite process constraints [17, 26] are positivity, $\mathrm{Tr}\, W = d_{A_O} d_{B_O} = 4$, and

$$
\begin{aligned}
{}_{B_I B_O}W &= {}_{A_O B_I B_O}W, & {}_{A_I A_O}W &= {}_{A_I A_O B_O}W, \\
W &= {}_{A_O}W + {}_{B_O}W - {}_{A_O B_O}W. &&
\end{aligned}
\tag{S3}
$$

The notation $A \prec B$ denotes Alice before Bob and $B \prec A$ the reverse order. An $A \prec B$ process additionally obeys ${}_{B_O}W = W$; a $B \prec A$ process obeys ${}_{A_O}W = W$. Causal separability (CS) means a sum of positive, possibly subnormalized, processes of these orders.

A normalized conditional distribution is causal if

$$
p = q\, p^{A \prec B} + (1-q)\, p^{B \prec A}, \qquad 0 \le q \le 1, \tag{S4}
$$

where p abbreviates the whole conditional table, the two superscripted tables are normalized distributions for the indicated orders, and q is the probability of the Alice-first branch. It is independent of x, y. Subscripts A, B on a distribution denote its Alice and Bob marginals: $p_A^{A \prec B}$ is independent of y, and $p_B^{B \prec A}$ independent of x. A process is extensibly causal (EC) if this holds for every finite shared input state and every finite collection of local instruments, settings, and outcomes [26]. The decomposition may depend on the experiment. Extensible causal separability is a stronger operator property [30, 31]; it is not the premise of our theorem.

Purifiability here has the general higher-order meaning [19]: the original process is obtained from a pure process with an added finite global past P and future F, by inserting a fixed pure state in P and tracing F. Their dimensions are unrestricted and are not identified with $\mathrm{rank}\, W$. A pure process preserves unitarity: inserting local unitary operations, including their local ancillary extensions, yields a global unitary transformation.

S2 Outline of the proof

Table S1 summarizes the reductions from arbitrary finite laboratory experiments to the operator inequality used below. These reductions preserve the Born probabilities and instrument normalization. The exact identity and positive weights then establish the bound in every finite representation. Numerical tests with complex instruments check the conventions, while the analytic reductions establish the full scope of the result.

S3 Physicality and an exact witness

The process in the main text is

$$
W_* = \frac{1}{4}\left[I + \frac{1}{5}\sigma_z^{A_I} + \frac{3}{10}\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O} + \frac{1}{5}\sum_{k=x,y,z} \sigma_k^{A_O}\sigma_k^{B_I} \right], \tag{S5}
$$

where k labels the three Pauli axes, and the asterisk labels the chosen counterexample. Every non-identity term lies in the linear subspace (S3); all are traceless, so $\mathrm{Tr}\, W_* = 4$. To check positivity, first

Experimental object	Representation	Preserved quantities and scope
Complex CP instruments	Full-transpose Choi convention and the positive conditional output	Original Born probabilities and the σ_y sign; Eqs. (S2), (S19), and (S24).
Input and output flags	CP refinement of the reports	Normalization and the original table after coarse graining; no assumption of diagonal original instruments.
Multiple Kraus branches	Rectangular Stinespring amplitudes with retained environment	Mixed event outputs and all quadratic pairings; zero padding adds no orthogonality relation; Eq. (S30).
General Bob effects	Fixed local Naimark input embedding	The original input qubit, every Pauli pairing, the joint probability, and the Hall cost; Eq. (S31).
All setting pairs	Common first-party marginals and a fixed order weight	A causal model for the complete experiment, including every outcome subset; Eq. (S26) and the transportation lemma.
Arbitrary finite auxiliary spaces	An operator sum of squares (SOS) modulo projector identities	Positivity in every finite representation; only positive Alice slack, Bob projector relations, and cross-laboratory commutation; Eq. (S45).
Mixed shared states	Convex combination or a local purification	A common order weight and arbitrary finite ancillas; Section S9.

Table S1: Physical coverage of the dimension-independent proof. The original laboratory ports remain qubits throughout.

fix the eigenvalues $s = \pm 1$ of $\sigma_z^{A_I}$ and $j = \pm 1$ of $\sigma_z^{B_O}$. On $A_O B_I$, $\sigma_x \otimes \sigma_x$ is conserved. In its sector $t = \pm 1$, the remaining eigenvalues are

$$\lambda_{s,t,\pm} = \frac{1 + s\delta + t\beta \pm \sqrt{\alpha^2 + \beta^2(1-t)^2}}{4}, \quad (\delta, \alpha, \beta) = \left(\frac{1}{5}, \frac{3}{10}, \frac{1}{5}\right), \tag{S6}$$

where δ, α, β are the coefficients of $\sigma_z^{A_I}$, $\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O}$, and the three-term Pauli sum, respectively; $\lambda_{s,t,\pm}$ labels the two eigenvalue branches in each sector. They do not depend on j. The full spectrum, with multiplicities in parentheses, is

$$\frac{1}{40}(2), \quad \frac{1}{8}(2), \quad \frac{7}{40}(2), \quad \frac{11}{40}(4), \quad \frac{13}{40}(2), \quad \frac{3}{8}(2), \quad \frac{17}{40}(2). \tag{S7}$$

Consequently W_* is a full-rank valid process and its smallest eigenvalue is $1/40$.

Let $\Pi_- = [I - \sum_{k=x,y,z} \sigma_k \otimes \sigma_k]/4$ be the singlet projector on $A_O B_I$. Define the Hermitian causal witness S by

$$S = (I - \sigma_z^{A_I})\Pi_- + \frac{1}{4}(I - \sigma_z^{A_I})\sigma_x^{B_I}\sigma_z^{B_O}. \tag{S8}$$

Its replaced operators are

$$\begin{aligned} {}_{B_O}S &= (I - \sigma_z^{A_I})\Pi_- \succeq 0, \\ {}_{A_O}S &= \frac{1}{4}(I - \sigma_z^{A_I})(I + \sigma_x^{B_I}\sigma_z^{B_O}) \succeq 0. \end{aligned} \tag{S9}$$

For an $A \prec B$ summand $W^{A\prec B}$, self-adjointness of the replacement map gives $\mathrm{Tr}(SW^{A\prec B}) = \mathrm{Tr}[({}_{B_O}S)W^{A\prec B}] \geq 0$; the other order uses ${}_{A_O}S$. Thus S is a CS witness. Pauli orthogonality gives exactly

$$\mathrm{Tr}(SW_*) = -\frac{1}{10}, \qquad \mathrm{Tr}(SI/4) = 1. \tag{S10}$$

S3.1 Relation to the existing transpose construction

The ordinary causal model follows from the construction of Feix, Araújo, and Brukner [25]. In the original Bob port bases,

$$\begin{aligned} W_*^{T_{B_I B_O}} &= A_T + B_T, \\ A_T &= \frac{1}{20}\left[I + \sigma_x^{A_O}\sigma_x^{B_I} - \sigma_y^{A_O}\sigma_y^{B_I} + \sigma_z^{A_O}\sigma_z^{B_I}\right], \\ B_T &= \frac{1}{4}\left[\frac{4}{5}I + \frac{1}{5}\sigma_z^{A_I} + \frac{3}{10}\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O}\right], \end{aligned} \tag{S11}$$

where $T_{B_I B_O}$ transposes only Bob's original input and output factors; the subscript T on A_T, B_T labels the components of this transposed process. A_T is a positive $A \prec B$ component with trace $4/5$ and nonzero eigenvalue $1/5$. B_T is a positive $B \prec A$ component with trace $16/5$ and minimum eigenvalue $3/40$. Both satisfy all process linear constraints. Bob's full Choi transpose preserves positive local instruments in the unextended experiment, giving an ordinary causal model for W_*.

This argument does not establish EC. Transposing an entire extended Bob instrument also transfers a partial transpose to the shared input state, which need not stay positive; transposing only the original ports need not preserve instrument positivity. Section VI and footnote 6 of Ref. [25] explicitly identify the gap between this construction, their finite searches, and extensible causality. The proof below supplies the missing arbitrary-ancilla statement.

S3.2 Operational response protocols

The laboratory responses discussed in the main text follow directly from the Born rule, with no ancillary state required. The axis label $k \in \{x, y, z\}$ selects the Pauli matrix σ_k, independently of the setting label x. Let $\rho_x^k = (I + x\sigma_k)/2$ be its eigenstate of eigenvalue $x = \pm 1$. Alice discards her input and prepares ρ_x^k; Bob measures σ_k, records $b = \pm 1$, and returns a maximally mixed qubit. In the full-transpose Choi convention the instruments are

$$M_x = I_{A_I} \otimes (\rho_x^k)^T, \qquad N_b = \frac{I + b\sigma_k}{2} \otimes \frac{I_{B_O}}{2}, \tag{S12}$$

where M_x represents Alice's single-outcome preparation channel at setting x and N_b Bob's measurement event with outcome b. The resulting distribution is

$$p(b|x,k) = \frac{1 + \kappa_k x b/5}{2}, \qquad \kappa_x = \kappa_z = 1, \quad \kappa_y = -1, \tag{S13}$$

where κ_k is the axis-dependent sign introduced by transposing the prepared state. The main-text protocol uses $k = x$. The σ_y sign is checked separately to detect a Choi convention error.

For the reverse response, Bob measures $\sigma_x^{B_I}$ with result $r = \pm 1$ and prepares a $\sigma_z^{B_O}$ eigenstate of eigenvalue yr, with setting $y = \pm 1$. Alice measures $\sigma_z^{A_I}$ with result $s = \pm 1$ and returns a maximally mixed qubit. Thus

$$M_s = \frac{I + s\sigma_z}{2} \otimes \frac{I}{2}, \qquad N_{r|y} = \frac{I + r\sigma_x}{2} \otimes \frac{I + yr\sigma_z}{2}, \tag{S14}$$

where M_s represents Alice's event with measured result s, and $N_{r|y}$ Bob's measurement and preparation event with result r and setting y; the tensor factors follow the laboratory input–output order. The joint distribution is $p(s, r|y) = [1 + s/5 + 3sy/10]/4$. Summing over r gives

$$\langle b \rangle_x = \frac{x}{5} \quad (k = x), \qquad \langle s \rangle_y = \frac{1}{5} + \frac{3y}{10}, \tag{S15}$$

where angle brackets denote the average recorded outcome conditional on the subscripted setting. Every displayed event is CP and each complete instrument is TP. These two separate experiments illustrate the operator's directional dependences; they do not supply its universal EC proof or a circuit realization. The independent verifier `repro/independent_operational_response_audit.py` forms the standard Choi matrices from the physical maps, applies the full transpose, and checks all probabilities and a deliberately incorrect σ_y transpose.

S4 Why nonseparability excludes every finite purification

The pure bipartite process structure theorem [27, 28] applies to a pure process whose unitary maps PA_OB_O to A_IB_IF. It gives orthogonal decompositions of its past and future spaces into an $A \prec B$ sector and a $B \prec A$ sector. The unitary restricted to each sector is an ordered pure comb. This includes degenerate or zero sectors.

Insert any normalized pure past state $|p\rangle = |p_{A\prec B}\rangle + |p_{B\prec A}\rangle$, where the subscripts identify its components in the two ordered past sectors. Although these components may be coherent, their images lie in orthogonal future sectors. Tracing F removes their cross terms. The reduced process is therefore

$$W = W^{A\prec B} + W^{B\prec A}, \qquad W^{A\prec B}, W^{B\prec A} \succeq 0, \tag{S16}$$

where $W^{A\prec B}, W^{B\prec A}$ are the subnormalized ordered process components, with weights $\|p_{A\prec B}\|^2$ and $\|p_{B\prec A}\|^2$; these are the squared Hilbert-space norms of the past components. No rank condition or bound on the added environment enters this argument.

Conversely, a finite CS process has ordered-comb Stinespring dilations of its two normalized branches. Purify each branch's source and retain every channel environment. Padding the branch environments permits a controlled direct sum of the two ordered pure dilations. Prepare a coherent pure past state with amplitudes equal to the square roots of the branch weights and retain an orthogonal order flag in the future. Discarding that future yields the original CS process. Therefore, for the present scope of trivial *original* past and future,

$$\text{Purifiable} = \text{CS}. \tag{S17}$$

The equality concerns sets of processes within the stated scope. The witness (S10) excludes W_* from both sets. It remains to prove EC.

The same exclusion rules out a standard deterministic quantum circuit with quantum control of causal order (QC-QC). Every such circuit admits a unitary extension by retaining its internal environments [39]. Preparing its fixed global input and discarding its future would constitute the purification excluded above. This statement concerns the standard circuit class in which each external operation is used once; postselected simulations and different implementation frameworks are outside its scope.

S5 Refinement and the complex Born reduction

S5.1 Refining the two classical flags

W_* commutes with $\sigma_z^{A_I}$ and $\sigma_z^{B_O}$. Dephasing these ports in their σ_z eigenbases leaves every Born probability unchanged. Refine Alice's report by its input flag $s = \pm 1$ and Bob's report by its output flag $j = \pm 1$. Sandwiching a Choi matrix with a flag projector preserves CP; summing refined reports preserves channel normalization. A causal model for the refined reports remains causal after their flags are discarded. We may hence prove EC for the refined experiment.

For a pure shared state choose Schmidt bases and write

$$|\psi\rangle = \sum_i \sqrt{\rho_i}|i\rangle_{A'}|i\rangle_{B'}, \qquad \rho = \sum_i \rho_i |i\rangle\langle i|, \qquad D = \sqrt{\rho}, \tag{S18}$$

where $|\psi\rangle$ is the shared pure state, ρ_i its nonnegative Schmidt probabilities summing to one, ρ the reduced density operator on Bob's paired Schmidt support, and D its positive square root. The indices i, j below label the paired ancillary basis vectors; $o, p \in \{0, 1\}$ label Alice's output-qubit basis. Unused coordinates can be padded with zeros. An accepted subset of Alice's reports in input sector s has Choi matrix M_s on $A'A_O$. Its conditional physical CP output on $A_O B'$ has entries

$$(\omega_s)_{oi,pj} = D_i D_j (M_s^T)_{io,jp}, \qquad \tau_s = \mathrm{Tr}_{A_O}\, \omega_s \preceq \rho, \qquad \omega_s \succeq 0, \tag{S19}$$

where ω_s is the subnormalized joint output and τ_s its Bob marginal; $D_i = \sqrt{\rho_i}$ is the corresponding diagonal entry of D. The transpose in M_s^T is the full transpose from Eq. (S2), followed by reordering to $A_O B'$. In particular,

$$\tau_s = D(\mathrm{Tr}_{A_O}\, M_s)^T D \preceq D^2 = \rho. \tag{S20}$$

ω_s itself is the untransposed physical CP output and can be mixed, with any finite number of Kraus branches. Write $\tau_{a,s|x}$ for the marginal associated with report a, sector s, and setting x. For the complete set of reports in either input sector, $\sum_a \tau_{a,s|x} = \rho$.

Bob's accepted subset in output sector j gives an effect E_j on $B_I B'$. Channel normalization implies

$$E_+, E_- \succeq 0, \qquad E = E_+ + E_- \preceq I_{B_I B'}, \tag{S21}$$

where E is the total effect of the accepted Bob subset, and E_+, E_- its two output-flag components. For all refined reports together their effects sum to $I_{B_I B'}$.

S5.2 The effective Bob operator

Under the identification of the A_O and B_I qubit bases, put $\omega_{s,k} = \mathrm{Tr}_{A_O}[(\sigma_k \otimes I)\omega_s]$ for $k = x, y, z$, where $\omega_{s,k}$ is the Pauli contraction of the joint output ω_s. The process contraction contains σ_k^T, as follows directly from Eq. (S19):

$$K_{s,j} = \frac{1 + s\delta}{4} I \otimes \tau_s + \frac{\beta}{4} \sum_{k=x,y,z} \sigma_k^T \otimes \omega_{s,k} + \frac{sj\alpha}{4} \sigma_x \otimes \tau_s, \tag{S22}$$

where $K_{s,j}$ is the effective operator on Bob's input for Alice sector s and Bob flag j. The Pauli sum equals $2\omega_s^{T_{A_O}} - I \otimes \tau_s$, where T_{A_O} denotes partial transpose on Alice's output factor only. For a qubit,

$$\omega_s^{T_{A_O}} = I \otimes \tau_s - (\sigma_y \otimes I)\omega_s(\sigma_y \otimes I). \tag{S23}$$

Combining these identities gives

$$\begin{aligned} K_{s,j} &= a_s I \otimes \tau_s - \frac{\beta}{2}(\sigma_y \otimes I)\omega_s(\sigma_y \otimes I) + \frac{sj\alpha}{4}\sigma_x \otimes \tau_s, \\ a_+ &= \frac{7}{20}, \qquad a_- = \frac{1}{4}, \end{aligned} \tag{S24}$$

where a_s is the flag-dependent scalar coefficient of $I \otimes \tau_s$. The probability of the selected pair of subsets is $\sum_{s,j} \mathrm{Tr}(E_j K_{s,j})$. This derivation fixes the σ_y sign and the Choi conjugations for complex instruments. An extra transpose of ω_s, of the shared state, or of a Bob block changes the contraction.

S6 The transportation criterion for all event subsets

Choose the same constants for every experiment:

$$q = \frac{49}{80}, \quad \theta = \frac{3}{25}, \quad w_s = \frac{q + s\theta}{2}, \quad w_+ = \frac{293}{800}, \quad w_- = \frac{197}{800}, \quad c = \frac{1-q}{2} = \frac{31}{160}, \tag{S25}$$

where q is the order weight, θ adjusts Alice's relative flag weights w_s, and c is the common coefficient of Bob's reference contribution. For refined reports a and b, with the port flags included in the reports, prescribe

$$g_a(x) = w_s \operatorname{Tr} \tau_{a,s|x}, \qquad h_b(y) = c \operatorname{Tr}[(I_{B_I} \otimes \rho) E_{b|y}], \tag{S26}$$

where $g_a(x), h_b(y)$ are the prescribed subnormalized first-party marginals in the two ordered branches and $E_{b|y}$ is Bob's refined event effect. These marginals are nonnegative and depend only on their own party's setting. Their total masses are $\sum_a g_a = q$ and $\sum_b h_b = 2c = 1 - q$.

Lemma 1 (Capacitated transportation)**.** *For a normalized nonnegative table p_{ab} and nonnegative g_a, h_b of masses $q, 1-q$, respectively, there are nonnegative tables u, v such that $p = u+v$, $\sum_b u_{ab} = g_a$, and $\sum_a v_{ab} = h_b$ if and only if*

$$p(\mathcal{A}, \mathcal{B}) \le g(\mathcal{A}) + h(\mathcal{B}) \quad \textit{for every pair of outcome subsets } \mathcal{A}, \mathcal{B}. \tag{S27}$$

In this criterion, p_{ab} is the joint table at a fixed setting pair. $p(\mathcal{A}, \mathcal{B})$ sums its entries over $a \in \mathcal{A}, b \in \mathcal{B}$, while $g(\mathcal{A}), h(\mathcal{B})$ sum the prescribed marginals over their respective subsets. The tables u, v are the subnormalized ordered contributions.

Proof. Necessity follows from $u(\mathcal{A}, \mathcal{B}) \le g(\mathcal{A})$ and $v(\mathcal{A}, \mathcal{B}) \le h(\mathcal{B})$. For sufficiency, including full and complementary subsets in Eq. (S27) gives $p_A(a) \ge g_a$ and $p_B(b) \ge h_b$, where p_A, p_B are the row and column marginals of p. Let $k_b = p_B(b) - h_b \ge 0$ be the residual column demand, of total mass q. A network with source–row capacities g_a, row–column capacities p_{ab}, and column–sink capacities k_b has a flow of mass q exactly when each cut has capacity at least q. If its source-side rows and columns are $\mathcal{S}, \mathcal{T}$, that condition is

$$\begin{aligned} &g(\mathcal{S}^c) + p(\mathcal{S}, \mathcal{T}^c) + k(\mathcal{T}) - q \\ &\qquad = g(\mathcal{S}^c) + h(\mathcal{T}^c) - p(\mathcal{S}^c, \mathcal{T}^c) \ge 0, \end{aligned} \tag{S28}$$

where the superscript c denotes the complement within the corresponding outcome set and $k(\mathcal{T})$ sums the residual demands over $\mathcal{T}$. These are precisely the assumed inequalities. The flow entries define $0 \le u_{ab} \le p_{ab}$, with row sums g_a and column sums k_b; setting $v = p - u$ proves the claim. □

Apply the lemma independently to every setting pair (x, y). The same prescribed $g_a(x)$ and $h_b(y)$ are used throughout. Then u/q is normalized and has Alice marginal $g_a(x)/q$, independent of y. Likewise $v/(1-q)$ has Bob marginal $h_b(y)/(1-q)$, independent of x. The setting-independent mixture weight is the fixed q. Thus proving every inequality (S27) proves causality of the whole experiment, regardless of the numbers of settings or outcomes.

Using the Born reduction, the left side minus the right side of an arbitrary such inequality is

$$H = \sum_{s,j} \operatorname{Tr}(E_j K_{s,j}) - \sum_s w_s \operatorname{Tr} \tau_s - c \operatorname{Tr}[(I \otimes \rho) E], \tag{S29}$$

where H is the scalar Hall excess for the selected event subsets. Sections S7–S9 prove $H \le 0$ for all allowed events and states, including empty, full, and complementary subsets.

S7 Lifting general CP events and Bob effects

S7.1 Retaining arbitrary Kraus environments

Write Alice's CP event in sector s with Kraus matrices $K_{s,\mu} : \mathcal{H}_A \to \mathbb{C}^2$, where $\mathcal{H}_A$ is her finite ancillary input space, $\mathbb{C}^2$ her qubit output space, and μ the Kraus-branch label. They satisfy $\sum_\mu K_{s,\mu}^\dagger K_{s,\mu} \preceq I_A$, with I_A the identity on $\mathcal{H}_A$. Retain their environment and set

$$L_{s,o} = \sum_\mu |\mu\rangle\langle o| K_{s,\mu}, \qquad R_s = I_A - \sum_{o=0}^{1} L_{s,o}^\dagger L_{s,o} \succeq 0, \tag{S30}$$

where $|\mu\rangle$ is an orthonormal basis of the retained environment $\mathcal{H}_E$, $\langle o|$ selects output basis state o, and R_s is the positive slack left by the accepted event. The rectangular amplitudes $L_{s,o} : \mathcal{H}_A \to \mathcal{H}_E$ reproduce ω_s after the environment is traced. Zero Kraus matrices pad the two environments to a common finite space. To interpret a square-matrix polynomial literally, embed $\mathcal{H}_A$ and $\mathcal{H}_E$ in a common larger Alice space and pad each $L_{s,o}$ with zero rows and columns. The shared state is embedded in the original input subspace. This preserves all $L_{s,o}^\dagger L_{s,p}$ matrix elements and leaves $I - \sum_o L_{s,o}^\dagger L_{s,o} \succeq 0$ on the whole enlarged space. Therefore arbitrary CP events, including mixed outputs and multiple Kraus branches, are covered by the contraction constraints.

S7.2 A dilation preserving the input-qubit operators

Set $E_0 = I - E_+ - E_-$ for the rejected effect. This three-outcome positive operator-valued measure (POVM) has a finite Naimark dilation. It can be chosen with the fixed input embedding

$$J = I_{B_I} \otimes I_{B'} \otimes |0\rangle, \qquad J^\dagger P_j J = E_j, \qquad P_j P_\ell = \delta_{j\ell} P_j, \tag{S31}$$

where J appends a fixed ancillary basis state $|0\rangle$, P_j are the dilated event projectors, and $j, \ell \in \{+, -, 0\}$ label the three outcomes. The Kronecker delta $\delta_{j\ell}$ is one for equal labels and zero otherwise; $\sum_j P_j = I$. To construct the projectors, let $|j\rangle$ be the three orthonormal ancillary outcome vectors. The isometry $V_N = \sum_j \sqrt{E_j} \otimes |j\rangle$ extends to a unitary U_N with $U_N J = V_N$; take $P_j = U_N^\dagger (I \otimes |j\rangle\langle j|) U_N$.

The added input space belongs to Bob's ancilla; B_I remains dimension two. The embedding J intertwines every B_I Pauli matrix. Thus Eq. (S31) preserves the I, σ_x, and σ_y terms of Eq. (S24), as well as the cost $\mathrm{Tr}[(I \otimes \rho) E_j]$. It preserves H exactly. We can hence replace Bob's effects by P_j and the shared state by its fixed local embedding.

S7.3 The Hall operator

For the dilated operators define the unnormalized qubit partial traces and blocks

$$T = \mathrm{Tr}_{B_I}(P_+ + P_-), \qquad V = \mathrm{Tr}_{B_I}[\sigma_x (P_+ - P_-)], \qquad F = \sigma_y (P_+ + P_-) \sigma_y, \tag{S32}$$

where T is the total accepted effect traced over the input qubit, V its σ_x-weighted outcome difference, and F the σ_y-conjugated accepted effect on the qubit and enlarged ancilla. In this subsection F is an operator, distinct from the global future space in Section S4. The Pauli matrices in Eq. (S32) act on B_I, with identities on Bob's enlarged ancilla. The (o, p) qubit block is denoted by F_{op}. All Alice operators commute with all Bob operators. Define the Hall operator by

$$\begin{aligned} \mathcal{H} = & \sum_{s,o} L_{s,o}^\dagger L_{s,o} (a_s T + s\alpha V/4 - w_s I) \\ & - \frac{\beta}{2} \sum_{s,o,p} L_{s,o}^\dagger L_{s,p} F_{op} - cT \end{aligned} \tag{S33}$$

where $\mathcal{H}$ acts on the two enlarged auxiliary spaces and its shared-state expectation is exactly the scalar H. In particular, the indices in its second line are F_{op}: expanding $\mathrm{Tr}(F\omega_s)$ gives $L^\dagger_{s,o}L_{s,p}F_{op}$. No transpose of this block occurs.

S8 A complete rational SOS identity

All the following matrices have operator-valued entries on the two auxiliary spaces, and their sizes are independent of those spaces' dimensions. For a real symmetric scalar matrix N and a compatible operator matrix Q, define

$$\langle N, Q\rangle = \sum_{i,j} N_{ij}Q_{ij}, \tag{S34}$$

where the indices i, j run over the common matrix size; the angle brackets denote a weighted matrix contraction, rather than the outcome average of Section S3.2. For $N, Q \succeq 0$ this contraction is positive. It is the scalar-index partial trace of $N^{1/2}QN^{1/2}$.

S8.1 Positive matrices and their bases

Write $C = P_0 = I - P_+ - P_-$ for the rejected projector, with qubit row and column indices $r, c \in \{0, 1\}$ and blocks C_{rc}. Use row-major ordering throughout and define the operator tuples

$$\begin{aligned} d &= (C_{00}, C_{01}, C_{10}, C_{11}),\\ n &= (C_{00} - C_{11}, C_{01}, C_{10}),\\ e &= ((P_+)_{00}, (P_+)_{01}, (P_+)_{10}, (P_+)_{11}, (P_-)_{00}, (P_-)_{01}, (P_-)_{10}, (P_-)_{11}),\\ v_s &= (L_{s,0}d, L_{s,1}d). \end{aligned} \tag{S35}$$

Products of an amplitude with a tuple act on each entry, and parentheses concatenate the resulting tuples. For an operator tuple $z = (z_i)$ define the Gram and slack-localized Gram matrices by

$$G(z)_{ij} = z_i^\dagger z_j, \qquad \mathcal{L}_{R_s}(z)_{ij} = z_i^\dagger R_s z_j, \tag{S36}$$

where $G(z)$ is the Gram matrix and $\mathcal{L}_{R_s}(z)$ inserts the positive Alice slack between tuple entries. Both are positive. The remaining two types are

$$(\mathcal{M}_s)_{(o,r),(p,c)} = L^\dagger_{s,o}C_{rc}L_{s,p}, \qquad (D_{s,j})_{rc} = R_s(P_j)_{rc}, \tag{S37}$$

where $\mathcal{M}_s$ is the accepted-amplitude localizer of the rejected projector and $D_{s,j}$ the slack–projector product matrix. Positivity of $\mathcal{M}_s$ follows by applying $C \succeq 0$ to $\sum_o L_{s,o}z_{o,r}$ for arbitrary test vectors $z_{o,r}$. Positivity of $D_{s,j}$ follows because R_s and P_j act on different laboratories. Compress $\mathcal{M}_s$ to

$$\widetilde{\mathcal{M}}_s = B^T\mathcal{M}_sB, \qquad B = \begin{pmatrix} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1 \end{pmatrix}, \tag{S38}$$

where B is the displayed real 4×3 compression matrix, $\widetilde{\mathcal{M}}_s$ the compressed localizer, and the four rows correspond to $(o, r) = (0, 0), (0, 1), (1, 0), (1, 1)$.

The eight-entry bases split under a reflection. For v_s choose representative and partner indices $(0, 1, 2, 3)$ and $(7, 6, 5, 4)$; for e choose $(0, 1, 4, 5)$ and $(3, 2, 7, 6)$. Indices start at zero. Define four-entry tuples

$$u_s^{\mathrm{e/o}} = (v_s)_{\mathrm{rep}} \pm (v_s)_{\mathrm{partner}}, \qquad z^{\mathrm{e/o}} = e_{\mathrm{rep}} \pm e_{\mathrm{partner}}, \tag{S39}$$

where "rep" and "partner" extract entries in the ordered index lists just specified; superscripts e, o denote even and odd reflection parity. The even tuple uses + and the odd tuple uses −. There is no square-root normalization factor.

S8.2 The fifteen scalar contractions

Let $\kappa = 4\,000\,000$ be the common weight denominator and $k_+ = 1, k_- = 8$ the starting matrix indices for the two flag sectors. The fifteen integer coefficient matrices $N_0, \dots, N_{14}$ listed below define the unscaled positive certificate operator $\mathcal{Q}$ by

$$\begin{aligned}\mathcal{Q} = {} & \langle N_0, G(n)\rangle \\ & + \sum_{s=+,-} \Big[\langle N_{k_s}, G(u_s^{\mathrm{e}})\rangle + \langle N_{k_s+1}, G(u_s^{\mathrm{o}})\rangle \\ & \quad + \langle N_{k_s+2}, \mathcal{L}_{R_s}(z^{\mathrm{e}})\rangle + \langle N_{k_s+3}, \mathcal{L}_{R_s}(z^{\mathrm{o}})\rangle \\ & \quad + \langle N_{k_s+4}, \widetilde{\mathcal{M}}_s\rangle + \langle N_{k_s+5}, D_{s,+}\rangle + \langle N_{k_s+6}, D_{s,-}\rangle\Big]. \end{aligned} \tag{S40}$$

Their actual SOS weights are N_k/κ. The explicit matrices are:

$$N_0 = \begin{pmatrix} 3368 & 0 & 0 \\ 0 & 3370 & 0 \\ 0 & 0 & 3370 \end{pmatrix}$$

$$N_1 = \begin{pmatrix} 376525 & 0 & 0 & 40190 \\ 0 & 164200 & 0 & 178330 \\ 0 & 0 & 336335 & 0 \\ 40190 & 178330 & 0 & 204390 \end{pmatrix}$$

$$N_2 = \begin{pmatrix} 376525 & 0 & 0 & 40190 \\ 0 & 164200 & 0 & -178330 \\ 0 & 0 & 336335 & 0 \\ 40190 & -178330 & 0 & 204390 \end{pmatrix}$$

$$N_3 = \begin{pmatrix} 191947 & 109881 & 80380 & 0 \\ 109881 & 111567 & 0 & 0 \\ 80380 & 0 & 191947 & -109881 \\ 0 & 0 & -109881 & 111567 \end{pmatrix}$$

$$N_4 = \begin{pmatrix} 111567 & 109881 & 0 & 0 \\ 109881 & 111567 & 0 & 0 \\ 0 & 0 & 111567 & -109881 \\ 0 & 0 & -109881 & 111567 \end{pmatrix}$$

$$N_5 = \begin{pmatrix} 99070 & 0 & 0 \\ 0 & 43340 & 0 \\ 0 & 0 & 99070 \end{pmatrix}$$

$$N_6 = \begin{pmatrix} 83606 & 80238 \\ 80238 & 83606 \end{pmatrix}$$

$$N_7 = \begin{pmatrix} 83606 & -80238 \\ -80238 & 83606 \end{pmatrix}$$

$$N_8 = \begin{pmatrix} 383070 & 0 & 0 & -38506 \\ 0 & 223259 & 0 & 198314 \\ 0 & 0 & 423259 & 0 \\ -38506 & 198314 & 0 & 183070 \end{pmatrix}$$

$$N_9 = \begin{pmatrix} 383070 & 0 & 0 & -38506 \\ 0 & 223259 & 0 & -198314 \\ 0 & 0 & 423259 & 0 \\ -38506 & -198314 & 0 & 183070 \end{pmatrix}$$

$$N_{10} = \begin{pmatrix} 110330 & -108517 & 113 & 0 \\ -108517 & 189025 & 0 & 78808 \\ 113 & 0 & 110330 & 108517 \\ 0 & 78808 & 108517 & 189025 \end{pmatrix}$$

$$N_{11} = \begin{pmatrix} 187342 & -108517 & 77125 & 0 \\ -108517 & 189025 & 0 & 78808 \\ 77125 & 0 & 187342 & 108517 \\ 0 & 78808 & 108517 & 189025 \end{pmatrix}$$

$$N_{12} = \begin{pmatrix} 3372 & 0 & 0 \\ 0 & 3372 & 0 \\ 0 & 0 & 3372 \end{pmatrix}$$

$$N_{13} = \begin{pmatrix} 86840 & -82966 \\ -82966 & 86840 \end{pmatrix}$$

$$N_{14} = \begin{pmatrix} 86840 & 82966 \\ 82966 & 86840 \end{pmatrix}$$

The matrix pairing and bases in Eqs. (S35)–(S40) specify each scalar coefficient. For example, N_1, N_2 weight Alice's $s = +$ forward Gram tuples, N_3, N_4 the same sector's slack-localized Bob tuples, N_5 its compressed complement localizer, and N_6, N_7 its two slack products. N_8 through N_{14} have the corresponding roles for $s = -$.

S8.3 All projection-ideal multipliers

The only equalities used in the identity are Bob's projector block relations. For $j, \ell \in \{+, -\}$ and $r, c \in \{0, 1\}$, denote the relation residuals by

$$\mathfrak{r}_{rc}^{j\ell} = \sum_{t=0}^{1} (P_j)_{rt}(P_\ell)_{tc} - \delta_{j\ell}(P_j)_{rc} = 0. \tag{S41}$$

There are sixteen such block polynomials $\mathfrak{r}_{rc}^{j\ell}$, all zero for the physical projectors. Their scalar and Alice-quadratic multiples give all 144 coefficients of the certificate. Define the ideal contribution $\mathcal{I}$ and its operator multipliers $B_{rc}^{j\ell}$ by

$$\begin{aligned} \mathcal{I} &= \sum_{j,\ell,r,c} B_{rc}^{j\ell}\, \mathfrak{r}_{rc}^{j\ell}, \\ B_{rc}^{j\ell} &= b_{rc}^{j\ell} I + \sum_{s,o,p} (C_s^{j\ell})_{(o,r),(p,c)} L_{s,o}^\dagger L_{s,p}, \end{aligned} \tag{S42}$$

where $b_{rc}^{j\ell}$ are scalar coefficients and $C_s^{j\ell}$ their 4×4 coefficient matrices for Alice's quadratic amplitudes; all entries are integers. Write $b^{j\ell}$ for the 2×2 matrix with entries $b_{rc}^{j\ell}$, and introduce its integer scalar coefficients a_0, b_0, c_0:

$$b^{++} = a_0 I + b_0 \sigma_x, \qquad b^{--} = a_0 I - b_0 \sigma_x, \qquad b^{+-} = b^{-+} = c_0 I. \tag{S43}$$

For each s, rows and columns of $C_s^{j\ell}$ are ordered $(o, r) = (0, 0), (0, 1), (1, 0), (1, 1)$.

Let $Z_4 = \mathrm{diag}(1,-1,-1,1)$ be the diagonal reflection matrix. The numerical coefficient matrices U_s, V_s determine the four label combinations through

$$C_s^{++} = U_s, \quad C_s^{+-} = V_s, \quad C_s^{--} = Z_4 U_s Z_4, \quad C_s^{-+} = V_s^T = Z_4 V_s Z_4. \tag{S44}$$

The constants and four integer matrices are:

$$a_0 = -604554, \qquad b_0 = -2728, \qquad c_0 = -160986.$$

$$U_+ = \begin{pmatrix} -449536 & 219762 & 0 & 0 \\ 219762 & -105266 & -356660 & 0 \\ 0 & -356660 & -105266 & 219762 \\ 0 & 0 & 219762 & -449536 \end{pmatrix}$$

$$V_+ = \begin{pmatrix} -672670 & 0 & 0 & 0 \\ 0 & -328400 & -356660 & 0 \\ 0 & -356660 & -328400 & 0 \\ 0 & 0 & 0 & -672670 \end{pmatrix}$$

$$U_- = \begin{pmatrix} -468468 & -217034 & 0 & 0 \\ -217034 & -68468 & -396628 & 0 \\ 0 & -396628 & -68468 & -217034 \\ 0 & 0 & -217034 & -468468 \end{pmatrix}$$

$$V_- = \begin{pmatrix} -688902 & 0 & 0 & 0 \\ 0 & -288902 & -396628 & 0 \\ 0 & -396628 & -288902 & 0 \\ 0 & 0 & 0 & -688902 \end{pmatrix}$$

Equations (S43)–(S44) thus print every ideal multiplier without leaving a free coefficient. Dividing $B_{rc}^{j\ell}$ by κ gives the multipliers for the weights N_k/κ.

S8.4 The exact identity and positivity

With the rational parameters (S25), direct expansion gives the polynomial identity

$$\boxed{-\kappa\mathcal{H} = \mathcal{Q} + \mathcal{I}.} \tag{S45}$$

The term $\mathcal{I}$ vanishes under the physical projector relations, while $\mathcal{Q}$ is positive. The identity holds in free Alice and Bob words with interparty commutation and the usual adjoint rule; within each party the multiplication order is retained. Every N_k is real symmetric and strictly positive definite. Table S2 gives their exact leading principal determinants $\Delta_r(N_k)$, with r the principal-block size. In an LDL factorization, L is unit lower triangular and D diagonal; the positive pivots d_r are successive determinant ratios. Consequently $\mathcal{Q} \succeq 0$ and $\mathcal{H} \preceq 0$.

The eleven-block certificate before reflection splitting is also supplied. Its integer coefficient system has size 656×318, rank 271, and 47 free coordinates. Rational recovery satisfies all coefficient equations exactly, and exact LDL factorizations certify all eleven weights. Averaging under joint A_O/B_I Pauli σ_x, then under joint Pauli σ_z with Bob's $+/-$ reports interchanged, preserves the Hall polynomial and positivity. Splitting the four eight-dimensional blocks yields precisely the fifteen matrices printed here. This transformation is algebraic and uses no further numerical optimization.

No Alice common-range, orthogonality, or nilpotent relation is used in Eq. (S45). Its only Alice assumption is $R_s \succeq 0$. The identity is therefore valid for arbitrary finite auxiliary dimensions. Taking its expectation in the embedded shared state proves $H \leq 0$.

k	$\Delta_1(N_k)$	$\Delta_2(N_k)$	$\Delta_3(N_k)$	$\Delta_4(N_k)$
0	3368	11350160	38250039200	–
1	376525	61825405000	20794047590675000	133585367050938762500
2	376525	61825405000	20794047590675000	133585367050938762500
3	191947	9341116788	1072171160231436	6835824856695625344
4	111567	373361328	41654803280976	139398681245923584
5	99070	4293693800	425376244766000	–
6	83606	551826592	–	–
7	83606	551826592	–	–
8	383070	85523825130	36198728700698670	110162193384172888504
9	383070	85523825130	36198728700698670	110162193384172888504
10	110330	9079188961	1001704504406905	7039978207917249424
11	187342	23636382261	3303716215774637	308260026331361563600
12	3372	11370384	38340934848	–
13	86840	657828444	–	–
14	86840	657828444	–	–

Table S2: Exact leading principal determinants of the fifteen integer SOS matrices N_k. The actual weights are $N_k/4000000$, so each rth leading determinant is $\Delta_r(N_k)/4000000^r$. All displayed integers are strictly positive; Sylvester's criterion proves positive definiteness. A dash denotes a principal block larger than the matrix. The LDL pivots are $d_r = \Delta_r/\Delta_{r-1}$ with $\Delta_0 = 1$.

S9 Universal EC and an independent conic route

The transportation lemma gives a useful sufficient criterion beyond the particular coefficients of W_*. For each shared state, prescribe nonnegative first-party marginals $g_a(x)$ and $h_b(y)$ of masses q and $1-q$, where q is independent of the settings. If every physical event-subset pair has Hall excess at most zero for all settings, the entire experiment is causal. An operator SOS valid in all physical representations is one way of proving these bounds simultaneously. The transportation criterion is general; the kernel, marginal choice, and rational weights supplied here certify the explicit example.

The arbitrary-dimensional inequality applies to the two Alice event subsets associated with any $\mathcal{A}$, and the two Bob event effects associated with any $\mathcal{B}$. Thus it proves every cut in Eq. (S27). The transportation lemma proves a causal decomposition for every pure shared state and every finite instrument collection. For a mixed shared state, decompose it into pure states and combine their causal models; the weight q in Eq. (S25) is common to all components. Equivalently, purify the state into an enlarged local input ancilla. Refinement and coarse graining then prove

$$W_* \in \mathrm{EC} \setminus \mathrm{CS}, \qquad W_* \notin \text{Purifiable}, \tag{S46}$$

where $\mathrm{EC} \setminus \mathrm{CS}$ means the set of EC processes excluding CS processes, $\in$ denotes membership, and $\notin$ nonmembership.

A second exact certificate is retained for the original, larger canonical conic relaxation. Its exact dual equations and zero objective provide an independent route to the same Hall bound. For completeness, the additional physical representation used by that relaxation can be attained without restricting CP events. Complete each accepted event by rejected Kraus matrices, pad the accepted slots for both sectors to the same number a, and pad the total environment to dimension m. Let Q_E project onto these a accepted slots. The complete Stinespring isometry $\mathcal{V}_s : \mathcal{H}_A \to \mathbb{C}^2 \otimes \mathbb{C}^m$ extends to a unitary $\mathcal{U}_s$ on a space of dimension $2m$, with a common input embedding J_A and $\mathcal{U}_s J_A = \mathcal{V}_s$. For any common isometry $F_E : \mathbb{C}^m \to \mathbb{C}^{2m}$ set

$$D_{s,o} = F_E Q_E (\langle o| \otimes I_m)\mathcal{U}_s, \qquad Q = F_E Q_E F_E^\dagger, \tag{S47}$$

where $D_{s,o}$ are dilated accepted-event amplitudes, Q their common output-range projector, and I_m the identity on the m-dimensional environment. These matrices are distinct from the Schmidt square root D and the slack products $D_{s,j}$ above. Then $D_{s,o}D_{s,p}^\dagger = \delta_{op}Q$ and $\sum_o D_{s,o}^\dagger D_{s,o} = \mathcal{U}_s^\dagger(I_2 \otimes Q_E)\mathcal{U}_s$

is a projector. Tracing the retained output environment recovers every original accepted CP output, including its mixedness. The Bob marginal and all Born pairings are preserved.

One further doubling gives

$$A_{s,o} = \begin{pmatrix} 0 & 0 \\ D_{s,o} & 0 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} J_A \\ 0 \end{pmatrix}, \tag{S48}$$

where $A_{s,o}$ are the amplitudes embedded in the doubled space and J_2 its input embedding. This preserves the common output range and enforces $A_{s,o}A_{t,p} = 0$ and $A_{s,o}^\dagger J_2 = 0$ for all sector labels s, t and output-basis indices o, p. This gives the state-supported 45-word Gram basis of the canonical relaxation. These relations apply to this representation, not to arbitrary original event matrices. The small certificate of Section S8 does not need this construction.

The independent rational verifier checks 10 960 canonical dual equations, an exact upper bound of zero, and positivity of the supported cone blocks. It also reconstructs the small SOS identity from the original Pauli contraction: the target polynomial contains 48 nonzero words, and the coefficient residual vanishes identically. The finite-dimensional proof above extends the Hall bound to arbitrary finite-dimensional shared states and all local instruments and establishes all transportation cuts. The numerical examples check the conventions and implementation; the general result follows from this proof.

S10 Exact classification on the white-noise line

Let

$$W_\eta = (1-\eta)W_* + \eta I/4, \qquad 0 \le \eta \le 1, \tag{S49}$$

where η is the probability of replacing the counterexample by the uniform process $I/4$, and the subscript labels the resulting noisy process. EC is convex, and $I/4$ is CS and hence EC. Therefore W_η is EC everywhere on this line. The witness value is

$$\mathrm{Tr}(SW_\eta) = \frac{11\eta - 1}{10}. \tag{S50}$$

It excludes CS and purification for $0 \le \eta < 1/11$.

To prove exactness, set $v = 1-\eta$ for the remaining visibility and write the two ordered components

$$\begin{aligned} W_\eta^{A\prec B} &= \frac{1}{4}\left[\frac{3v}{5} I + \frac{v}{5}\sum_{k=x,y,z} \sigma_k^{A_O}\sigma_k^{B_I}\right], \\ W_\eta^{B\prec A} &= \frac{1}{4}\left[\left(1 - \frac{3v}{5}\right) I + \frac{v}{5}\sigma_z^{A_I} + \frac{3v}{10}\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O}\right]. \end{aligned} \tag{S51}$$

Their sum is W_η, and each obeys its ordered-process linear constraints. The first is positive with eigenvalues 0 and $v/5$ and weight $3v/5$. The second has weight $1 - 3v/5$ and minimum eigenvalue $(1 - 11v/10)/4$, which is nonnegative exactly when $\eta \ge 1/11$. Hence

$$W_\eta \in \mathrm{EC} \quad \text{for all } \eta, \qquad W_\eta \in \mathrm{Purifiable} \iff W_\eta \in \mathrm{CS} \iff \eta \ge \frac{1}{11}. \tag{S52}$$

The double arrows mean "if and only if." The boundary $\eta = 1/11$ is included in CS and purifiability. The line admits the following stability extension in the full valid-process space.

S11 A neighborhood stable in every valid perturbation direction

All norms and traces in this section use the process normalization $\operatorname{Tr} W = 4$. Let $\mathcal{V}_0$ be the Hermitian, trace-zero subspace obeying the homogeneous constraints (S3); its real dimension is 87. For a Hermitian perturbation D in this section, the operator norm $\|D\|_\infty$ is its largest absolute eigenvalue; this D is distinct from the Schmidt square root in Section S5. The trace norm $\|S\|_1$ is the sum of absolute eigenvalues of the witness.

Lemma 2 (An explicit CS inner ball)**.** *For every $D \in \mathcal{V}_0$,*

$$\|D\|_\infty \le \frac{1}{12} \quad \Longrightarrow \quad I/4 + D \in \mathrm{CS}. \tag{S53}$$

Proof. Put $R_A = {}_{A_O}$ and $R_B = {}_{B_O}$. These commuting output-replacement maps preserve the process linear subspace. They are positive and unital, so they contract the operator norm on Hermitian inputs: apply each map to $-\|D\|_\infty I \preceq D \preceq \|D\|_\infty I$. Equation (S3) gives $D = R_A D + R_B D - R_A R_B D$. Define

$$\begin{aligned} A(D) &= \frac{I}{8} + R_B D - \frac{1}{2} R_A R_B D, \\ B(D) &= \frac{I}{8} + R_A D - \frac{1}{2} R_A R_B D, \end{aligned} \tag{S54}$$

where $A(D), B(D)$ are candidate subnormalized components of orders $A \prec B, B \prec A$, respectively; juxtaposition of a map and an operator means map application. Their sum is $I/4 + D$. Each remains in the process linear subspace, has trace two, and obeys $R_B A(D) = A(D)$ or $R_A B(D) = B(D)$, respectively. Norm contraction and the triangle inequality imply

$$\lambda_{\min}(A(D)), \lambda_{\min}(B(D)) \ge \frac{1}{8} - \frac{3}{2}\|D\|_\infty \ge 0, \tag{S55}$$

where $\lambda_{\min}$ denotes the smallest eigenvalue. Thus these are positive ordered components, proving the lemma. □

Theorem 1 (A full-dimensional EC, nonpurifiable neighborhood)**.** *Let $W_0 = I/4$ and*

$$\overline{W} = \frac{21}{22} W_* + \frac{1}{22} W_0, \tag{S56}$$

where W_0 is the uniform process and $\overline{W}$ the center at noise fraction $1/22$. For every perturbation $\Delta \in \mathcal{V}_0$,

$$\|\Delta\|_\infty \le \frac{1}{264} \quad \Longrightarrow \quad \overline{W} + \Delta \in \mathrm{EC} \setminus \mathrm{CS}. \tag{S57}$$

Every process in this ball is nonpurifiable and has no standard deterministic QC-QC realization.

Proof. Set $D = 22\Delta$. The lemma gives $W_0 + D \in \mathrm{CS}$, and

$$\overline{W} + \Delta = \frac{21}{22} W_* + \frac{1}{22}(W_0 + D). \tag{S58}$$

Both normalized components are EC, so convexity proves EC and also ensures positivity and normalization of the perturbed process.

The witness (S8) has a positive first term of trace four. Its remaining term is $(I - \sigma_z^{A_I})\sigma_x^{B_I}\sigma_z^{B_O}/4$, of trace norm four. Hence $\|S\|_1 \le 8$ by the triangle inequality; its exact value is $2 + 2\sqrt{5}$. Using Eq. (S50) and trace-norm duality,

$$\begin{aligned} \operatorname{Tr}[S(\overline{W} + \Delta)] &\le -\frac{1}{20} + 8\|\Delta\|_\infty \\ &\le -\frac{1}{20} + \frac{8}{264} = -\frac{13}{660} < 0. \end{aligned} \tag{S59}$$

This excludes CS throughout the closed ball. Equation (S17) and the QC-QC implication above prove the remaining exclusions. □

More generally, for $0 < \eta < 1/11$ the same argument proves $W_\eta + \Delta \in \mathrm{EC} \setminus \mathrm{CS}$ whenever

$$\Delta \in \mathcal{V}_0, \qquad \|\Delta\|_\infty < \min\left\{\frac{\eta}{12}, \frac{1-11\eta}{80}\right\}. \tag{S60}$$

These are conservative sufficient radii, not exact boundaries. They establish a nonzero-volume region relative to the trace-fixed valid-process affine space; they do not establish that W_* itself is interior to EC.

For a parameter-space view, set

$$W(\delta,\alpha,\beta) = \frac{1}{4}\left[I + \delta\sigma_z^{A_I} + \alpha\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O} + \beta\sum_{k=x,y,z}\sigma_k^{A_O}\sigma_k^{B_I}\right], \tag{S61}$$

where δ, α, β are varied real coefficients of the same Pauli terms.

The center parameters are $(21/110, 63/220, 21/110)$. Write $d\delta, d\alpha, d\beta$ for finite offsets from this center, rather than infinitesimal differentials. Since the Pauli sum has norm three, the neighborhood includes all such offsets satisfying

$$|d\delta| + |d\alpha| + 3|d\beta| \le \frac{1}{66}. \tag{S62}$$

The full ball also includes valid directions outside this three-parameter family, including ones that break the original flag symmetries.

The conclusions concern finite original qubit ports and arbitrary finite shared-input ancillary assistance. Trusted-input and network certification use additional assumptions [37, 38]. No claim is made about all certification scenarios, infinite-dimensional processes, or multiple-use composition; a tensor product of mixed-order processes is not automatically a valid process for grouped laboratories.

S12 Reproduction and independent checks

Data availability.—The exact certificates and portable verification code are publicly available in the GitHub repository and in the accompanying supplemental files. They check the process, witness, laboratory responses, and stability without optimization.

The package supplies the original eleven-block certificate in `results/exact_double_flag_hall_sos.json`, its symmetry average in `results/exact_double_flag_hall_sos_symmetric.json`, and the integer tables, ideal multipliers, and principal determinants in the three corresponding `hall_sos_*` JSON files. The three TeX fragments used above are generated directly from those rational data.

From the project root, the following commands verify both small certificates using Python's standard-library `Fraction` arithmetic, independently of the optimization used to obtain them:

```
python -m repro.independent_general_ec_certificate_audit \
  --certificate results/exact_double_flag_hall_sos.json \
  --output results/independent_general_ec_certificate_audit.json
python -m repro.independent_general_ec_certificate_audit \
  --certificate results/exact_double_flag_hall_sos_symmetric.json \
  --output results/independent_symmetric_hall_sos_audit.json
python -m repro.independent_counterexample_process_audit
python -m repro.independent_ec_robustness_audit
```

The continuation syntax shown is for a POSIX shell; the same arguments can be supplied on a single line on Windows. Table regeneration and the independent conic audit are:

```
python -m repro.export_hall_sos_tables
python results/literature_scope/independent_two_certificate_ec_audit.py
python results/literature_scope/independent_ec_physical_coverage_audit.py
```

Regenerating the tables requires SymPy, NumPy, SciPy, and CVXPY. The calculation evaluates the supplied rational coefficients exactly, without further optimization. The verification reports equation residuals and the rational pivots or determinants that establish positivity.

Finite numerical tests with complex mixed states compare the original Born table with Eq. (S24), refine the flags, enumerate every subset cut, and solve the transportation problem. The tests include multiple Kraus branches per report and unequal ancillary dimensions. Inserting an additional transpose makes the two probability calculations disagree. The Born-reduction residuals are below 5×10^{-17} and the transportation residuals below 2×10^{-15}. These tests check the Choi convention and its implementation; the dimension-independent result follows from Eqs. (S45) and (S28).

The portable distribution provides a single entry point, `verify.py`, and requires only Python and SymPy. It checks the two exact SOS certificates, process validity, the witness, the ordered decomposition along every valid Pauli basis direction, and the neighborhood constants. All checks use the supplied rational data and require neither optimization nor certificate regeneration. The reductions and proofs in Sections S2–S11 establish coverage of arbitrary finite laboratory experiments; the pure-process structure theorem [27, 28] supplies the link between causal nonseparability and the exclusion of finite pure dilations.

References

[1] W. F. Stinespring, Positive functions on C^*-algebras, Proc. Am. Math. Soc. **6**, 211 (1955).

[2] G. Chiribella, G. M. D'Ariano, P. Perinotti, and B. Valiron, Quantum computations without definite causal structure, Phys. Rev. A **88**, 022318 (2013).

[3] C. Brukner, Quantum causality, Nat. Phys. **10**, 259 (2014).

[4] G. Chiribella, Perfect discrimination of no-signalling channels via quantum superposition of causal structures, Phys. Rev. A **86**, 040301(R) (2012).

[5] M. Araújo, F. Costa, and C. Brukner, Computational Advantage from Quantum-Controlled Ordering of Gates, Phys. Rev. Lett. **113**, 250402 (2014).

[6] N. Ma, P. Z. Zhao, and J. Gong, Quantum machine learning with indefinite causal order, Phys. Rev. A **110**, 052406 (2024).

[7] P. A. Guérin, A. Feix, M. Araújo, and C. Brukner, Exponential Communication Complexity Advantage from Quantum Superposition of the Direction of Communication, Phys. Rev. Lett. **117**, 100502 (2016).

[8] D. Ebler, S. Salek, and G. Chiribella, Enhanced Communication with the Assistance of Indefinite Causal Order, Phys. Rev. Lett. **120**, 120502 (2018).

[9] G. Chiribella *et al.*, Indefinite causal order enables perfect quantum communication with zero capacity channels, New J. Phys. **23**, 033039 (2021).

[10] L. M. Procopio *et al.*, Experimental superposition of orders of quantum gates, Nat. Commun. **6**, 7913 (2015).

[11] G. Rubino *et al.*, Experimental verification of an indefinite causal order, Sci. Adv. **3**, e1602589 (2017).

[12] K. Goswami *et al.*, Indefinite Causal Order in a Quantum Switch, Phys. Rev. Lett. **121**, 090503 (2018).

[13] K. Wei *et al.*, Experimental Quantum Switching for Exponentially Superior Quantum Communication Complexity, Phys. Rev. Lett. **122**, 120504 (2019).

[14] G. Chiribella, G. M. D'Ariano, and P. Perinotti, Transforming quantum operations: Quantum supermaps, Europhys. Lett. **83**, 30004 (2008).

[15] G. Chiribella, G. M. D'Ariano, and P. Perinotti, Quantum Circuit Architecture, Phys. Rev. Lett. **101**, 060401 (2008).

[16] G. Chiribella, G. M. D'Ariano, and P. Perinotti, Theoretical framework for quantum networks, Phys. Rev. A **80**, 022339 (2009).

[17] O. Oreshkov, F. Costa, and C. Brukner, Quantum correlations with no causal order, Nat. Commun. **3**, 1092 (2012).

[18] G. Chiribella, G. M. D'Ariano, and P. Perinotti, Probabilistic theories with purification, Phys. Rev. A **81**, 062348 (2010).

[19] M. Araújo, A. Feix, M. Navascués, and C. Brukner, A purification postulate for quantum mechanics with indefinite causal order, Quantum **1**, 10 (2017).

[20] M. Ueda and M. Kitagawa, Reversibility in quantum measurement processes, Phys. Rev. Lett. **68**, 3424 (1992).

[21] S. Hong, Y.-S. Kim, Y.-W. Cho, J. Kim, S.-W. Lee, and H.-T. Lim, Demonstration of Complete Information Trade-Off in Quantum Measurement, Phys. Rev. Lett. **128**, 050401 (2022).

[22] C. Branciard, M. Araújo, A. Feix, F. Costa, and C. Brukner, The simplest causal inequalities and their violation, New J. Phys. **18**, 013008 (2016).

[23] M. Araújo *et al.*, Witnessing causal nonseparability, New J. Phys. **17**, 102001 (2015).

[24] C. Branciard, Witnesses of causal nonseparability: an introduction and a few case studies, Sci. Rep. **6**, 26018 (2016).

[25] A. Feix, M. Araújo, and C. Brukner, Causally nonseparable processes admitting a causal model, New J. Phys. **18**, 083040 (2016).

[26] O. Oreshkov and C. Giarmatzi, Causal and causally separable processes, New J. Phys. **18**, 093020 (2016).

[27] W. Yokojima, M. T. Quintino, A. Soeda, and M. Murao, Consequences of preserving reversibility in quantum superchannels, Quantum **5**, 441 (2021).

[28] J. Barrett, R. Lorenz, and O. Oreshkov, Cyclic quantum causal models, Nat. Commun. **12**, 885 (2021).

[29] O. Oreshkov, Time-delocalized quantum subsystems and operations: on the existence of processes with indefinite causal structure in quantum mechanics, Quantum **3**, 206 (2019).

[30] J. Wechs, A. A. Abbott, and C. Branciard, On the definition and characterisation of multipartite causal (non)separability, New J. Phys. **21**, 013027 (2019).

[31] W. Wei and S. Pang, Extensibly Causally Separable Processes Admit Realizations as Quantum Circuits with Classical Control of Causal Order, arXiv:2609.18559v1 (2026).

[32] J. Wechs, A. A. Abbott, and C. Branciard, All causally separable quantum processes are quantum circuits with classical control of causal order, arXiv:2609.20774v1 (2026).

[33] L.-M. Duan, M. D. Lukin, J. I. Cirac, and P. Zoller, Long-distance quantum communication with atomic ensembles and linear optics, Nature **414**, 413 (2001).

[34] M. Chen, J.-S. Tang, M. Cai, Y. Lu, F. Nori, and K. Xia, High-dimensional two-photon quantum controlled phase-flip gate, Phys. Rev. Research **6**, 033004 (2024).

[35] X.-W. Luo, C. Zhang, I. Novikova, C. Qian, and S. Du, Wavelength conversion for single-photon polarization qubits through continuous-variable quantum teleportation, Phys. Rev. A **105**, 052444 (2022).

[36] See Supplemental Material for process physicality, the general Born reduction, the transportation theorem, all rational SOS matrices, and independent verification commands.

[37] H. Dourdent *et al.*, Semi-Device-Independent Certification of Causal Nonseparability with Trusted Quantum Inputs, Phys. Rev. Lett. **129**, 090402 (2022).

[38] H. Dourdent *et al.*, Network-Device-Independent Certification of Causal Nonseparability, Quantum **8**, 1514 (2024).

[39] J. Wechs, H. Dourdent, A. A. Abbott, and C. Branciard, Quantum circuits with classical versus quantum control of causal order, PRX Quantum **2**, 030335 (2021).

[40] S. Choi, Y. Bao, X.-L. Qi, and E. Altman, Quantum Error Correction in Scrambling Dynamics and Measurement-Induced Phase Transition, Phys. Rev. Lett. **125**, 030505 (2020).

[41] B. M. Garraway, Nonperturbative decay of an atomic system in a cavity, Phys. Rev. A **55**, 2290 (1997).

[42] T. Yu and J. H. Eberly, Sudden Death of Entanglement, Science **323**, 598 (2009).

[43] D. Fallas Padilla, V. So, A. Menon, R. Zhuravel, H. Pu, and G. Pagano, Delocalized Excitation Transfer in Open Quantum Systems with Long-Range Interactions, PRX Quantum **6**, 040301 (2025).

[44] C. Li, Y. Wu, and W.-M. Liu, Non-Hermitian superfluid–Mott-insulator transition in the one-dimensional zigzag bosonic chains, Phys. Rev. B **109**, 214306 (2024).